Category Theory for Mathematical Physics — Exercise Sheet 0

Diagnostic: what the course assumes

Variant: Solutions

This optional sheet tests the prerequisites; each exercise names the sibling lectures that teach them and the lectures of this course that use them. Vector spaces are finite-dimensional over $\mathbb{C}$, and in $g \circ f$ the map f acts first. For V with basis $e_1, \dots, e_n$ ($n = \dim V$), the dual basis of V^* is $e^1, \dots, e^n$, $e^i(e_j) = \delta_{ij}$; $m = \dim W$. P is the flip $v \otimes w \mapsto w \otimes v$, and the partial trace $\mathrm{tr}_2 \colon \mathrm{End}(V \otimes W) \to \mathrm{End}(V)$ is $\mathrm{tr}_2(M) = \sum_k (\mathrm{id} \otimes e^k)\, M\, (\mathrm{id} \otimes e_k)$ for a basis e_k of W. Groups are multiplicative, with identity 1. The singlet is written $|\psi^-\rangle$, and the antisymmetric tensor is $\Omega = e_1 \otimes e_2 - e_2 \otimes e_1$, as in Lecture 4.

Exercise 1 (Duals and the trace). (*Linear Algebra*, Lectures 11–12; used from Lecture 4.) Let $\mathrm{ev} \colon V^* \otimes V \to \mathbb{C}$, $\varphi \otimes v \mapsto \varphi(v)$, and $\mathrm{coev} \colon \mathbb{C} \to V \otimes V^*$, $1 \mapsto \sum_i e_i \otimes e^i$.

(i) Show that coev does not depend on the basis.

(ii) Show that $(\mathrm{id}_V \otimes \mathrm{ev}) \circ (\mathrm{coev} \otimes \mathrm{id}_V) = \mathrm{id}_V$ and $(\mathrm{ev} \otimes \mathrm{id}_{V^*}) \circ (\mathrm{id}_{V^*} \otimes \mathrm{coev}) = \mathrm{id}_{V^*}$.

(iii) Show that $\mathrm{ev} \circ P \circ (f \otimes \mathrm{id}_{V^*}) \circ \mathrm{coev} = \mathrm{tr}\, f$ for $f \colon V \to V$; in particular $\mathrm{ev} \circ P \circ \mathrm{coev} = \dim V$.

(iv) Show that $\varphi \otimes w \mapsto \big(v \mapsto \varphi(v)\, w\big)$ is an isomorphism $V^* \otimes W \to \mathrm{Hom}(V, W)$.

Solution. (i) With $f_k = \sum_i B_{ik} e_i$ and $f^k = \sum_j (B^{-1})_{kj} e^j$ (indeed $f^k(f_l) = \sum_i (B^{-1})_{ki} B_{il} = \delta_{kl}$),

$$\sum_k f_k \otimes f^k = \sum_{i,j} \Big(\sum_k B_{ik} (B^{-1})_{kj} \Big) e_i \otimes e^j = \sum_i e_i \otimes e^i .$$

(ii) $(\mathrm{id} \otimes \mathrm{ev})(\mathrm{coev} \otimes \mathrm{id})(e_j) = \sum_i e_i\, e^i(e_j) = e_j$, and $(\mathrm{ev} \otimes \mathrm{id})(\mathrm{id} \otimes \mathrm{coev})(\varphi) = \sum_i \varphi(e_i)\, e^i = \varphi$. These are the *snake identities* of Lecture 4, drawn there as a bent strand pulled straight.

(iii) $(f \otimes \mathrm{id})\mathrm{coev}(1) = \sum_i f e_i \otimes e^i$; P makes it $\sum_i e^i \otimes f e_i$, and ev gives $\sum_i e^i(f e_i) = \sum_i f_{ii} = \mathrm{tr}\, f$. For $f = \mathrm{id}$ this is $n = \dim V$, the closed loop of Lecture 4.

(iv) The map is linear in φ and w. Its image of $\sum_i e^i \otimes g(e_i)$ is $v \mapsto \sum_i e^i(v) g(e_i) = g(v)$, so it is onto, and both spaces have dimension nm; so it is an isomorphism, with that inverse. (For $W = V$ the element $P\,\mathrm{coev}(1) = \sum_i e^i \otimes e_i$ corresponds to id_V.)

Exercise 2 (Tensor products of maps). (*Linear Algebra*, Lecture 12; *Quantum Mechanics I*, Lectures 10–11; used from Lecture 3.)

(i) Show that $(f \otimes g) \circ (f' \otimes g') = f f' \otimes g g'$ and $P \circ (f \otimes g) \circ P = g \otimes f$ for $f \colon V \to V$ and $g \colon W \to W$, where the right P is the flip $W \otimes V \to V \otimes W$ and the left one $V \otimes W \to W \otimes V$.

(ii) On $\mathbb{C}^2 \otimes \mathbb{C}^2$, find the eigenvalues of P with their eigenvectors.

(iii) Show that P on $\mathbb{C}^2 \otimes \mathbb{C}^2$ is not of the form $f \otimes g$.

(iv) Show that $\mathrm{tr}_2(A \otimes B) = (\mathrm{tr}\, B)\, A$.

Solution. (i) $(f \otimes g)(f' \otimes g')(v \otimes w) = f(f'v) \otimes g(g'w)$, and $P(f \otimes g)P(w \otimes v) = P(fv \otimes gw) = gw \otimes fv = (g \otimes f)(w \otimes v)$; both sides are linear and agree on product vectors, which span.

(ii) $P^2 = 1$, so the eigenvalues are ± 1. The symmetric tensors $e_1 \otimes e_1$, $e_2 \otimes e_2$, $e_1 \otimes e_2 + e_2 \otimes e_1$ have eigenvalue $+1$ and the antisymmetric $e_1 \otimes e_2 - e_2 \otimes e_1$ has -1: eigenvalues $1, 1, 1, -1$.

(iii) Suppose $P = f \otimes g$. Then $f e_1 \otimes g e_1 = P(e_1 \otimes e_1) = e_1 \otimes e_1$, so $f e_1 = c e_1$ for some $c \neq 0$. Then $P(e_1 \otimes e_2) = c e_1 \otimes g e_2$ lies in $e_1 \otimes \mathbb{C}^2$, but $P(e_1 \otimes e_2) = e_2 \otimes e_1$ does not. So the flip, the most basic map of the course after the identity, is not a product of maps on the factors: it is

a new piece of structure (the tensor product is Lecture 3's, and the flip is the symmetry of Lecture 7).

(iv) By the definition in the Conventions, $\operatorname{tr}_2(A\otimes B) = \sum_k (\mathrm{id}\otimes e^k)(A\otimes B)(\mathrm{id}\otimes e_k) = \sum_k A\, e^k(Be_k) = (\operatorname{tr} B)\,A$. (*Quantum Mechanics I*, Lecture 11, defines the reduced state by the equivalent identity $\operatorname{Tr}(\rho_1 A) = \operatorname{Tr}(\rho(A\otimes 1))$.)

Exercise 3 (Characters of S_3). (*Symmetries in Physics*, Lectures 3–5 and 8; used from Lecture 2.) Let std be the permutation representation of S_3 on $\mathbb{C}^3$ minus its trivial summand.

(i) Write the characters of **1**, sgn and std on the three conjugacy classes, and check that they are orthonormal.

(ii) Decompose std ⊗ std.

(iii) Find $\dim \operatorname{Hom}_{S_3}(\mathrm{std}\otimes\mathrm{std},\mathrm{std})$, and show that every S_3-map std → std is a scalar.

Solution. (i) The classes are $\{1\}$, the three transpositions and the two 3-cycles. The permutation character is the number of fixed points, $(3,1,0)$, so $\chi_{\mathrm{std}} = (3,1,0) - (1,1,1) = (2,0,-1)$; $\chi_{\mathrm{sgn}} = (1,-1,1)$. With $\langle\chi,\psi\rangle = \frac{1}{6}\sum_g \chi(g)\overline{\psi(g)}$: $\langle\chi_{\mathrm{std}},\chi_{\mathrm{std}}\rangle = \frac{1}{6}(4+0+2) = 1$, $\langle\chi_{\mathrm{std}},\chi_{\mathbf{1}}\rangle = \frac{1}{6}(2+0-2) = 0$, $\langle\chi_{\mathrm{std}},\chi_{\mathrm{sgn}}\rangle = 0$, and the others likewise. So std is irreducible, and with $1^2+1^2+2^2 = 6 = |S_3|$ these are all the irreducible representations.

(ii) $\chi_{\mathrm{std}\otimes\mathrm{std}} = \chi_{\mathrm{std}}^2 = (4,0,1)$; its inner products with $\mathbf{1}, \mathrm{sgn}, \mathrm{std}$ are $\frac{1}{6}(4+0+2) = 1$, $\frac{1}{6}(4+0+2) = 1$, $\frac{1}{6}(8+0-2) = 1$. So $\mathrm{std}\otimes\mathrm{std} \cong \mathbf{1}\oplus\mathrm{sgn}\oplus\mathrm{std}$, of dimension 4.

(iii) By (ii) and Schur's lemma, $\dim \operatorname{Hom}_{S_3}(\mathrm{std}\otimes\mathrm{std},\mathrm{std}) = 1$. An S_3-map $T\colon \mathrm{std}\to\mathrm{std}$ has an eigenvalue λ, and $T-\lambda$ is an S_3-map with a nonzero kernel, an invariant subspace; by irreducibility the kernel is everything, so $T = \lambda$. These multiplicity spaces are the fusion spaces of Lecture 5.

Exercise 4 (Spin and Clebsch–Gordan). (*Symmetries in Physics*, Lectures 12–14; *Quantum Mechanics I*, Lecture 13; used from Lecture 2, and in Lectures 4 and 9.) Let $V_{1/2} = \mathbb{C}^2$ be the defining representation of SU(2), with basis $e_1 = \uparrow$, $e_2 = \downarrow$, and V_j the spin-j representation.

(i) Show that $V_{1/2}\otimes V_{1/2} \cong V_0 \oplus V_1$, with V_0 spanned by the singlet $(e_1\otimes e_2 - e_2\otimes e_1)/\sqrt{2}$, and that the singlet is fixed by $U\otimes U$ for every $U\in\mathrm{SU}(2)$.

(ii) With the characters $\chi_j(t) = \sin\big((2j+1)t/2\big)/\sin(t/2)$ on the rotation by t (extended continuously where the denominator vanishes), show that $\chi_{1/2}\chi_j = \chi_{j-1/2} + \chi_{j+1/2}$ for $j\geq\frac{1}{2}$, and deduce $V_{1/2}\otimes V_j \cong V_{j-1/2}\oplus V_{j+1/2}$.

(iii) Decompose $V_1\otimes V_1$ and check the dimensions.

Solution. (i) For $U = \left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right)$, $(U\otimes U)(e_1\otimes e_2 - e_2\otimes e_1) = (ae_1+ce_2)\otimes(be_1+de_2) - (be_1+de_2)\otimes(ae_1+ce_2) = (ad-bc)(e_1\otimes e_2 - e_2\otimes e_1)$, and $\det U = 1$: the singlet spans an invariant line, V_0. Its complement, the symmetric tensors (dimension 3), is invariant ($U\otimes U$ commutes with P) and contains the highest weight vector $e_1\otimes e_1$ of spin 1; so it is V_1, and $V_{1/2}\otimes V_{1/2}\cong V_0\oplus V_1$. (The total spin J^2 is 0 on the singlet and $2 = 1\cdot 2$ on the triplet.)

(ii) With $x = (2j+1)t/2$ and $\chi_{1/2}(t) = \sin t/\sin(t/2) = 2\cos(t/2)$: $2\cos(t/2)\sin x = \sin(x+\frac{t}{2}) + \sin(x-\frac{t}{2}) = \sin((2j+2)\frac{t}{2}) + \sin(2j\frac{t}{2})$, and dividing by $\sin(t/2)$ gives $\chi_{1/2}\chi_j = \chi_{j+1/2}+\chi_{j-1/2}$. Characters of SU(2) determine representations up to isomorphism, and the character of a tensor product is the product, so $V_{1/2}\otimes V_j \cong V_{j-1/2}\oplus V_{j+1/2}$. Truncating this rule at level k is how $\mathrm{SU}(2)_k$ arises (Lectures 5 and 9).

(iii) By (ii) at $j=\frac{1}{2}$, $\chi_1 = \chi_{1/2}^2 - \chi_0$. Then $\chi_1\chi_1 = \chi_{1/2}(\chi_{1/2}\chi_1) - \chi_1 = \chi_{1/2}(\chi_{1/2}+\chi_{3/2}) - \chi_1 = (\chi_0+\chi_1) + (\chi_1+\chi_2) - \chi_1 = \chi_0+\chi_1+\chi_2$, so $V_1\otimes V_1 \cong V_0\oplus V_1\oplus V_2$ (the Clebsch–Gordan range $|j-j'|,\dots,j+j'$), and $3\cdot 3 = 9 = 1+3+5$.

Exercise 5 (Two qubits). (*Quantum Mechanics I*, Lectures 10–11; used from Lecture 3.) Let $|\psi^-\rangle = (|01\rangle - |10\rangle)/\sqrt{2}$ in $\mathbb{C}^2\otimes\mathbb{C}^2$, with $|0\rangle = e_1$ and $|1\rangle = e_2$.

(i) Show that $|\psi^-\rangle$ is not a product $|a\rangle \otimes |b\rangle$.

(ii) Compute the reduced state $\rho_1 = \mathrm{tr}_2 |\psi^-\rangle\langle\psi^-|$, and compare with tr_2 of a product state.

(iii) How does P act on $|\psi^-\rangle$, and on $(|01\rangle + |10\rangle)/\sqrt{2}$?

Solution. (i) $|a\rangle\otimes|b\rangle$ has coefficient matrix $c_{ij} = a_i b_j$, of rank at most 1; $|\psi^-\rangle$ has $c = \frac{1}{\sqrt{2}}\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$, of rank 2. So $|\psi^-\rangle$ is entangled.

(ii) $\rho_1 = cc^\dagger = \frac{1}{2}\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) = \frac{1}{2}\cdot 1$: the first qubit alone is maximally mixed. For $|a\rangle \otimes |b\rangle$ with unit vectors, $\rho_1 = |a\rangle\langle a|\ \langle b|b\rangle = |a\rangle\langle a|$, a pure state, with $\rho_1^2 = \rho_1$.

(iii) $P|\psi^-\rangle = (|10\rangle - |01\rangle)/\sqrt{2} = -|\psi^-\rangle$, and P fixes $(|01\rangle + |10\rangle)/\sqrt{2}$: the singlet and the triplet of Exercise 4 are the eigenspaces of the exchange. The tensor product of systems is Lecture 3's monoidal structure, and P exchanging them is the symmetry of Lecture 7.

Exercise 6 ⋆. *An invariant cup with loop* -2 (a preview of Lecture 4). On $V = \mathbb{C}^2$ let $\Omega = e_1 \otimes e_2 - e_2 \otimes e_1 \in V \otimes V$, and let $\Omega^\dagger : V \otimes V \to \mathbb{C}$ be the linear map with $\Omega^\dagger(e_1 \otimes e_2) = 1$, $\Omega^\dagger(e_2 \otimes e_1) = -1$, $\Omega^\dagger(e_i \otimes e_i) = 0$.

(i) Show that Ω and $\Omega^\dagger$ are invariant under SU(2): $(U \otimes U)\Omega = \Omega$ and $\Omega^\dagger \circ (U \otimes U) = \Omega^\dagger$.

(ii) Compute $(\mathrm{id} \otimes \Omega^\dagger) \circ (\Omega \otimes \mathrm{id})$ and $(\Omega^\dagger \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \Omega)$ as maps $V \to V$.

(iii) Show that if an invariant cup $\lambda\Omega$ and an invariant cap $\mu\Omega^\dagger$ satisfy both snake identities of Exercise 1, then the loop $(\mu\Omega^\dagger)(\lambda\Omega)$ is -2. Why can no invariant cup and cap that satisfy both snakes make the loop $+2 = \dim V$, as in Exercise 1?

Solution. (i) Ω is invariant by Exercise 4(i). $\Omega^\dagger(x \otimes y) = x_1y_2 - x_2y_1 = \det(x\,y)$, the determinant of the matrix with columns x, y, so $\Omega^\dagger(Ux \otimes Uy) = \det(U(x\,y)) = \det U\, \Omega^\dagger(x \otimes y) = \Omega^\dagger(x \otimes y)$.

(ii) $(\mathrm{id} \otimes \Omega^\dagger)(\Omega \otimes v) = e_1\, \Omega^\dagger(e_2 \otimes v) - e_2\, \Omega^\dagger(e_1 \otimes v)$. For $v = e_1$: $-e_1$; for $v = e_2$: $-e_2$. So the first snake is $-\mathrm{id}$, and in the same way $(\Omega^\dagger \otimes \mathrm{id})(v \otimes \Omega) = \Omega^\dagger(v \otimes e_1)e_2 - \Omega^\dagger(v \otimes e_2)e_1 = -v$.

(iii) By Exercise 4(i) the invariant vectors of $V \otimes V$ form the line V_0, so an invariant cup is $\lambda\Omega$, and likewise an invariant cap is $\mu\Omega^\dagger$ (an invariant form on $V \otimes V$ vanishes on V_1). By (ii) the snakes are $-\lambda\mu\ \mathrm{id}$, so they equal id exactly when $\lambda\mu = -1$. Then the loop is $\lambda\mu\, \Omega^\dagger(\Omega) = -2$, whatever the choice. (Without the snakes the pair $(\Omega, \Omega^\dagger)$ itself has loop $+2$, but its snakes are $-\mathrm{id}$.) In Exercise 1 the loop is $+\dim V$, but there the cup lies in $V \otimes V^*$; an invariant cup in $V \otimes V$ identifies V with its dual through the antisymmetric form $\Omega^\dagger$, and the antisymmetry costs the sign (Theorem 4.8). Lecture 4 records this sign as the Frobenius–Schur indicator $\kappa_{1/2} = -1$, and Lecture 9 meets it again as the loop value $-[2]_q$ at $q = 1$.